

Medical Engineering

Gastro-Respiratory Monitoring System

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ABSTRACT

The assessment of gastrointestinal and respiratory activity in clinical practice is commonly performed using manual and intermittent methods, which are subjective and do not support continuous monitoring. This study presents the design, development, and testing of a wearable and remote gastro-respiratory monitoring system aimed at enabling real-time and continuous physiological observation.

The proposed system integrates piezoelectric and resistive sensing principles for the acquisition of gastrointestinal and respiratory signals, respectively. A signal modeling approach was first implemented in MATLAB, where realistic gastric and respiratory waveforms were generated with added noise to simulate real-world conditions. Digital filtering techniques were applied to isolate relevant frequency components, forming the basis for subsequent hardware design.

An embedded system based on the ESP32 microcontroller was developed to perform real-time data acquisition, processing, and wireless communication. Simulated signals from MATLAB were transmitted to the ESP32 via a Wi-Fi network, enabling validation of the system without reliance on physical sensors. The received signals were processed and displayed on a web-based dashboard hosted by the ESP32, providing real-time visualization of gastrointestinal and respiratory activity.

The results demonstrated successful separation of high-frequency gastric signals and low-frequency respiratory signals, accurate real-time data transmission, and stable system performance with minimal latency. The system also implemented a correlation analysis framework to explore the relationship between the two physiological signals.

The study confirms the feasibility of developing a low-cost, portable, and integrated gastro-respiratory monitoring system using embedded and wireless technologies. Although the system was validated using simulated data, it provides a strong foundation

for future development involving real physiological signal acquisition and clinical application.

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LIST OF ACRONYMS AND ABBREVIATION

1. GI – Gastrointestinal
2. ADC – Analog to Digital Converter
3. MCU – Microcontroller Unit
4. ESP32 – Espressif 32-bit Microcontroller
5. Wi-Fi – Wireless Fidelity
6. FFT – Fast Fourier Transform
7. GAI – Gastric Activity Index
8. RSI – Respiratory Stress Index
9. PCB – Printed Circuit Board
10. IoT – Internet of Things
11. GSM – Global System for Mobile Communication
12. BT – Bluetooth

CHAPTER 1: INTRODUCTION

1.0 Background information

Monitoring body sounds and movements is essential in today's medical diagnosis and patient care. Among the various physiological signals used in clinical settings, abdominal sounds and breathing patterns offer important insights into the gastrointestinal (GI) tract and respiratory system. Gastrointestinal sounds, or bowel sounds, occur when fluids and gases move within the intestines during digestion. Clinicians often use these sounds to assess intestinal activity and digestive health. Likewise, breathing patterns provide key information about how the lungs, diaphragm, and surrounding muscles are functioning. Abnormalities in these patterns can signal respiratory issues, metabolic problems, or complications from abdominal conditions.

Typically, clinicians check abdominal sounds with a stethoscope during a physical exam. This process, called auscultation, involves listening to bowel sounds in certain areas of the abdomen to see if intestinal activity is normal, increased, decreased, or absent. While this method has been used for many years because it is simple and inexpensive, it has notable drawbacks. First, auscultation only gives a brief view of the patient's condition at the time of the exam. It does not allow for continuous monitoring or long-term data collection. Second, interpreting abdominal sounds is very subjective and relies heavily on the clinician's experience and judgment. Consequently, two different clinicians may come to different conclusions about the same patient. Moreover, manual auscultation may miss subtle changes or patterns that happen over time.

In many gastrointestinal disorders, such as intestinal obstruction, ileus, inflammatory bowel disease, and post-operative complications, bowel sounds may change gradually. Continuous monitoring of these sounds could give clinicians useful information about the progress of the disease and recovery. However, without automated monitoring systems, such changes often go unnoticed until symptoms become severe. This issue highlights the need for innovative technologies that can continuously capture, analyze, and store gastrointestinal sound data for clinical use.

Monitoring breathing is another critical part of patient assessment, especially in those with abdominal or chest conditions. The respiratory system works closely with the abdominal muscles and diaphragm, so problems in the abdomen can affect breathing patterns. For instance, abdominal swelling, infection, trauma, or surgical complications can restrict normal diaphragm movement and breathing. Despite this link, gastrointestinal and respiratory monitoring are usually done separately in clinical practice. Combining both functions into one monitoring system could provide a better understanding of a patient's overall condition.

Recent advancements in biomedical engineering and embedded systems have made it possible to create wearable medical devices that can continuously monitor physiological signals. Wearable sensors allow for real-time data collection while keeping patients comfortable and mobile. These technologies are becoming increasingly important in modern healthcare, especially for remote patient monitoring and telemedicine. Remote monitoring systems help healthcare providers keep track of patients outside of hospital

settings, enabling early detection of medical issues and reducing the need for frequent hospital visits.

Among the many sensing technologies available, piezoelectric sensors are particularly good at detecting vibrations and sounds from the body. Piezoelectric materials generate electrical signals when subjected to pressure or vibrations, making them effective for capturing abdominal sounds related to intestinal activity. Similarly, resistive sensors can detect changes like chest or abdominal expansion during breathing. These sensors turn physical changes into measurable electrical signals, which can then be processed with microcontrollers.

Microcontroller platforms like Arduino are now popular in biomedical prototyping because they are flexible, affordable, and easy to connect with various sensors and communication tools. By combining sensors with microcontrollers and wireless communication technologies, it's possible to design systems that capture physiological signals, process them in real-time, and send the data to remote platforms for monitoring and analysis. Wireless technologies such as Wi-Fi, Bluetooth, and GSM allow healthcare professionals to access patient data from almost any location, supporting the growing fields of telemedicine and digital healthcare.

The combination of wearable sensing technologies and wireless communication offers new chances for continuous health monitoring. These systems can gather physiological data over long periods, identify patterns that may suggest the early stages of disease, and

send alerts when abnormal conditions arise. Additionally, long-term data collection can support clinical research by providing objective data that can be statistically analyzed.

This project suggests creating a Wearable and Remote Gastro-Respiratory Monitoring System that uses piezoelectric sensors and resistive respiratory sensors connected to an Arduino-based microcontroller. The goal is to capture abdominal sounds and breathing patterns at the same time, process the gathered information, and send it wirelessly to a remote monitoring platform. By bringing together multiple physiological parameters in one wearable device, this system will offer a broader assessment of a patient's digestive and respiratory health.

The device will act as a non-invasive, portable monitoring solution suitable for both hospital and home use. Patients will be able to wear the device comfortably while clinicians receive real-time updates via a remote monitoring interface. The system will also allow for long-term tracking of physiological signals, enabling healthcare providers to observe trends and identify issues that may not be visible during regular clinical exams.

Ultimately, developing such a system could improve diagnostic accuracy, support early detection of gastrointestinal and respiratory disorders, and enhance the overall quality of patient care. By utilizing modern sensor technology, embedded systems, and wireless communication, this project contributes to the ongoing development of wearable biomedical monitoring devices and remote healthcare solutions.

1.1 Statement of the problem

The current method for assessing gastrointestinal sounds relies on manual auscultation with a stethoscope. This approach is subjective and does not allow for continuous monitoring. There is also no effective system for remotely monitoring abdominal sounds and related respiratory issues. This makes it hard for clinicians to observe trends in digestive health over time. Without an automated system, diagnosis is delayed and management of gastrointestinal and lower abdominal respiratory disorders is insufficient.

1.2 Justification of study

Conventional assessment of abdominal sounds and breathing patterns is typically done through periodic clinical exams with a stethoscope. This method only gives brief observations and relies heavily on the clinician's experience. It makes it hard to notice subtle or gradual changes that could signal early stages of gastrointestinal or respiratory issues. By combining sensors that detect abdominal vibrations and respiratory movements with a microcontroller platform like the Arduino Uno, the proposed system allows for continuous, real-time monitoring of these physiological parameters. This system can improve early detection of problems, help with remote patient monitoring, and lessen the limitations of manual clinical assessments.

1.3 Objective

1.3.1 Main objective

The main objective of this study is to design, construct and test a wearable and remote gastro-respiratory monitoring system.

1.3.2 Specific objectives

The study had three specific objectives:

1. To design a wearable and remote gastro-respiratory monitoring system.
2. To construct a wearable and remote gastro-respiratory monitoring system using Piezo-electric and resistive sensors integrated with an Arduino-based microcontroller.
3. To test the wearable and remote gastro-respiratory monitoring system with simulated inputs to determine the viability of the monitoring system.

1.4 Scope of study

This study looks at creating a wearable gastro-respiratory monitoring system that can detect abdominal sounds and breathing movements in real time. The system will use vibration and motion sensing technologies combined with a microcontroller platform like the Arduino Uno to gather and process physiological signals from the body. The study will include choosing the right sensors, developing the hardware interface, programming the microcontroller, and testing the system to make sure it can reliably acquire signals. The project will mainly focus on developing a prototype and basic signal monitoring, rather than on complex clinical diagnosis or large medical trials. Its goal is to show how

combining gastro-intestinal sound detection and respiratory monitoring can work in one wearable device.

1.5 Significance of study

The importance of this study is in its potential to improve the monitoring of digestive and respiratory functions with a portable and non-invasive system. By allowing continuous monitoring of abdominal sounds and breathing patterns, the proposed device can help healthcare professionals identify problems earlier than traditional, occasional exams. The system may also support remote patient monitoring, enabling clinicians to observe patient conditions without needing constant hospital visits. Additionally, this project shows how low-cost embedded systems and sensor technologies can be used for medical monitoring. Therefore, this research could improve patient care, support early diagnosis of gastro-respiratory issues, and promote the development of affordable biomedical monitoring solutions.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of the Research Area

Gastrointestinal (GI) and respiratory monitoring are critical components of modern biomedical diagnostics, as both systems provide essential indicators of physiological health. Gastrointestinal activity, particularly bowel sounds, reflects digestive motility and has been widely studied as a non-invasive biomarker for disorders such as irritable bowel syndrome, intestinal obstruction, and postoperative ileus (Craine et al., 2002; Du et al., 2018). Traditionally, bowel sounds have been assessed using auscultation; however, this method is subjective and highly dependent on clinician expertise, leading to variability in diagnosis (Dudekula et al., 2011).

Similarly, respiratory monitoring plays a vital role in detecting and managing conditions such as chronic obstructive pulmonary disease (COPD), asthma, and sleep apnea. Continuous respiratory rate monitoring is considered a key vital sign and an early indicator of clinical deterioration (Subbe et al., 2001). Conventional respiratory monitoring systems, such as spirometers and capnographs, though accurate, are often bulky and unsuitable for continuous, real-world use (Massaroni et al., 2019).

The emergence of wearable biomedical monitoring systems has significantly transformed healthcare delivery by enabling continuous, real-time physiological data acquisition outside clinical settings (Pantelopoulos & Bourbakis, 2010). Wearable sensors allow for long-term monitoring, early detection of abnormalities, and improved patient outcomes. In recent years, advancements in flexible electronics, microcontrollers such as Arduino,

and low-power sensors have facilitated the development of compact and cost-effective monitoring devices (Heikenfeld et al., 2018).

Despite these advancements, integrated systems capable of simultaneously monitoring gastrointestinal and respiratory signals remain underexplored. Most existing systems focus on either respiratory or gastrointestinal parameters independently. This gap highlights the need for a unified wearable gastro-respiratory monitoring system that leverages multiple sensing modalities, such as piezoelectric and resistive sensors, to provide comprehensive physiological insights.

2.2 Review of Previous Studies

Bowel Sound Detection

Research on bowel sound analysis has evolved significantly with the application of digital signal processing techniques. Early work by Craine et al. (2002) demonstrated the feasibility of automated bowel sound analysis using acoustic sensors, highlighting the potential for objective GI monitoring. Subsequent studies by Du et al. (2018) employed machine learning algorithms to classify bowel sounds, achieving improved accuracy in detecting gastrointestinal motility patterns.

Yin et al. (2017) further advanced this field by developing a wearable bowel sound monitoring system that utilized MEMS microphones and signal filtering techniques. Their findings emphasized the importance of noise reduction and feature extraction in improving signal quality. Similarly, Spiegel et al. (2014) demonstrated that continuous

bowel sound monitoring could be used as a non-invasive biomarker for digestive health, reinforcing its clinical relevance.

However, challenges remain in distinguishing bowel sounds from environmental noise and motion artifacts, which limits the reliability of existing systems (Dudekula et al., 2011).

Respiratory Monitoring Using Wearable Sensors

Wearable respiratory monitoring has been widely investigated using various sensing modalities. Massaroni et al. (2019) provided a comprehensive review of wearable systems for respiratory monitoring, highlighting technologies such as strain sensors, piezoelectric sensors, and impedance-based systems. Their study emphasized the importance of comfort, accuracy, and long-term usability in wearable designs.

Nam et al. (2014) developed a flexible piezoelectric sensor for respiratory monitoring, demonstrating high sensitivity and the ability to detect subtle chest movements. Similarly, Yoon et al. (2014) introduced a stretchable strain sensor capable of accurately measuring respiration by detecting thoracic expansion.

Resistive sensors, particularly those based on conductive fabrics or polymers, have also been widely used. Atalay et al. (2017) developed a textile-based resistive sensor integrated into clothing for respiration monitoring, showing promising results in terms of comfort and signal fidelity. Mattmann et al. (2008) explored piezoresistive strain

sensors for wearable applications, demonstrating their effectiveness in detecting body movements and respiratory patterns.

Piezoelectric Sensors in Biomedical Applications

Piezoelectric sensors have gained significant attention due to their ability to convert mechanical stress into electrical signals. Wang (2006) highlighted the potential of piezoelectric nanogenerators for biomedical sensing applications, emphasizing their sensitivity and energy efficiency.

In respiratory monitoring, piezoelectric sensors have been used to detect chest wall movements with high precision (Nam et al., 2014). Additionally, piezoelectric transducers have been applied in acoustic sensing for bowel sound detection, as they are capable of capturing low-frequency vibrations associated with gastrointestinal activity (Yin et al., 2017).

Dagdeviren et al. (2014) demonstrated the use of flexible piezoelectric devices for biomedical monitoring, showing their potential for integration into wearable systems. These sensors offer advantages such as high sensitivity, low power consumption, and the ability to detect both acoustic and mechanical signals.

Resistive/Strain Sensors for Respiration Monitoring

Resistive sensors operate based on changes in electrical resistance due to mechanical deformation. They are widely used in wearable systems due to their simplicity and

flexibility. Mattmann et al. (2008) demonstrated that piezoresistive sensors could effectively measure respiratory movements by detecting strain in elastic materials.

Textile-based resistive sensors have been particularly promising for wearable applications. Atalay et al. (2017) showed that conductive yarns integrated into garments could provide accurate respiratory measurements while maintaining user comfort. Similarly, Stoppa and Chiolerio (2014) highlighted the role of wearable electronics in enabling continuous physiological monitoring.

Comparatively, resistive sensors are generally more cost-effective and easier to integrate than piezoelectric sensors, but they may suffer from issues such as signal drift and lower sensitivity (Massaroni et al., 2019).

Comparative Analysis

Across the literature, piezoelectric sensors are often preferred for high-sensitivity applications, particularly where vibration detection is required, such as bowel sounds. In contrast, resistive sensors are more suitable for detecting large-scale mechanical movements, such as chest expansion during respiration. Studies suggest that combining these sensing modalities could provide complementary data, improving overall system performance (Massaroni et al., 2019; Atalay et al., 2017).

2.3 Existing Technologies or Systems

Several technologies have been developed for physiological monitoring. Electronic stethoscopes, for instance, enhance traditional auscultation by amplifying and digitally

processing acoustic signals (Leng et al., 2015). These devices are commonly used for detecting heart and lung sounds, and some have been adapted for bowel sound analysis.

Wearable respiratory monitoring systems include chest belts equipped with strain gauges or piezoelectric sensors. Commercial devices such as respiratory inductance plethysmography systems measure thoracic and abdominal movements to estimate respiratory rate (Massaroni et al., 2019).

IoT-based healthcare systems have also gained prominence, enabling remote monitoring through wireless communication. Pantelopoulos and Bourbakis (2010) discussed the integration of wearable sensors with wireless networks, allowing real-time data transmission to healthcare providers.

Microcontroller platforms such as Arduino have been widely used in prototype development due to their affordability and ease of use (Banzi & Shiloh, 2014). These platforms enable the integration of multiple sensors into a single system, facilitating real-time data acquisition and processing.

2.4 Limitations of Existing Systems

Despite significant advancements, existing monitoring systems exhibit several limitations. One major issue is the lack of integration between gastrointestinal and respiratory monitoring systems. Most devices are designed to measure a single physiological parameter, limiting their ability to provide comprehensive health insights (Massaroni et al., 2019).

Cost is another significant barrier, particularly for advanced monitoring systems that rely on specialized sensors and proprietary technologies (Heikenfeld et al., 2018). Additionally, many systems are not designed for continuous, long-term use due to discomfort, battery limitations, or data storage constraints.

Signal quality remains a critical challenge, especially in wearable systems where motion artifacts and environmental noise can degrade measurements (Yin et al., 2017). Furthermore, resistive sensors may experience drift over time, affecting accuracy (Mattmann et al., 2008).

Another limitation is the lack of real-time data processing and user-friendly interfaces in many existing systems, which restricts their practical application in home-based healthcare settings.

2.5 Research Gap

The literature reveals a clear gap in the development of integrated wearable systems capable of simultaneously monitoring gastrointestinal and respiratory signals. While significant progress has been made in individual domains, few studies have explored multi-modal systems that combine acoustic and mechanical sensing techniques.

Additionally, there is limited research on low-cost, Arduino-based systems that integrate piezoelectric and resistive sensors for real-time monitoring. Most existing systems rely on expensive hardware, limiting accessibility, particularly in resource-constrained settings.

The need for a compact, wearable, and cost-effective gastro-respiratory monitoring system is therefore evident. Such a system could provide continuous, non-invasive monitoring, enabling early detection of abnormalities and improving patient outcomes.

2.6 Theoretical/Conceptual Framework

The proposed system is grounded in three key principles: piezoelectric sensing, resistive sensing, and physiological signal acquisition.

Piezoelectric sensing is based on the generation of electrical charge in response to mechanical stress. Materials such as polyvinylidene fluoride (PVDF) exhibit piezoelectric properties, making them suitable for detecting vibrations and acoustic signals (Wang, 2006). In this project, piezoelectric sensors are used to capture bowel sounds and subtle respiratory vibrations.

Resistive sensing, on the other hand, relies on changes in electrical resistance due to deformation. When a resistive sensor is stretched or compressed, its resistance changes proportionally, allowing for the measurement of physical movement (Mattmann et al., 2008). This principle is applied to detect chest expansion during respiration.

Physiological signal acquisition involves capturing, amplifying, and processing biological signals. This typically includes filtering noise, digitizing signals using analog-to-digital converters, and analyzing data using microcontrollers (Webster, 2009). Arduino serves as the central processing unit in the proposed system, enabling real-time data acquisition and analysis.

The integration of these principles allows for the development of a multi-modal monitoring system capable of capturing both gastrointestinal and respiratory signals, providing a comprehensive assessment of physiological function.

2.7 Summary of Literature Review

This literature review has examined the current state of research in gastrointestinal and respiratory monitoring, with a focus on wearable systems and sensor technologies. The findings indicate that while significant progress has been made in individual monitoring techniques, there is a lack of integrated systems capable of simultaneously capturing multiple physiological signals.

Piezoelectric and resistive sensors have been shown to be effective for detecting bowel sounds and respiratory movements, respectively. However, existing systems often suffer from limitations such as high cost, lack of portability, and susceptibility to noise and signal drift.

The identified research gap highlights the need for a low-cost, wearable gastro-respiratory monitoring system that integrates multiple sensing modalities. The proposed system addresses this gap by leveraging piezoelectric and resistive sensors in conjunction with Arduino-based signal processing, offering a promising solution for continuous, non-invasive health monitoring.

3.0 RESEARCH DESIGN AND METHODOLOGY

3.1 Overview of Methodological Framework

The methodology for the design and development of the gastro-respiratory monitoring system was structured as a sequential engineering workflow consisting of system-level design, signal modeling, circuit analysis, algorithm validation, and hardware implementation. The approach integrates MATLAB-based simulation with analog circuit design and embedded system deployment using ESP32.

The methodology emphasizes quantitative parameter selection, reproducible simulation environments, and direct mapping between simulated and real-time signals to ensure consistency across all development stages.

3.2 System Design Approach

The system design began with the formulation of a functional block diagram,

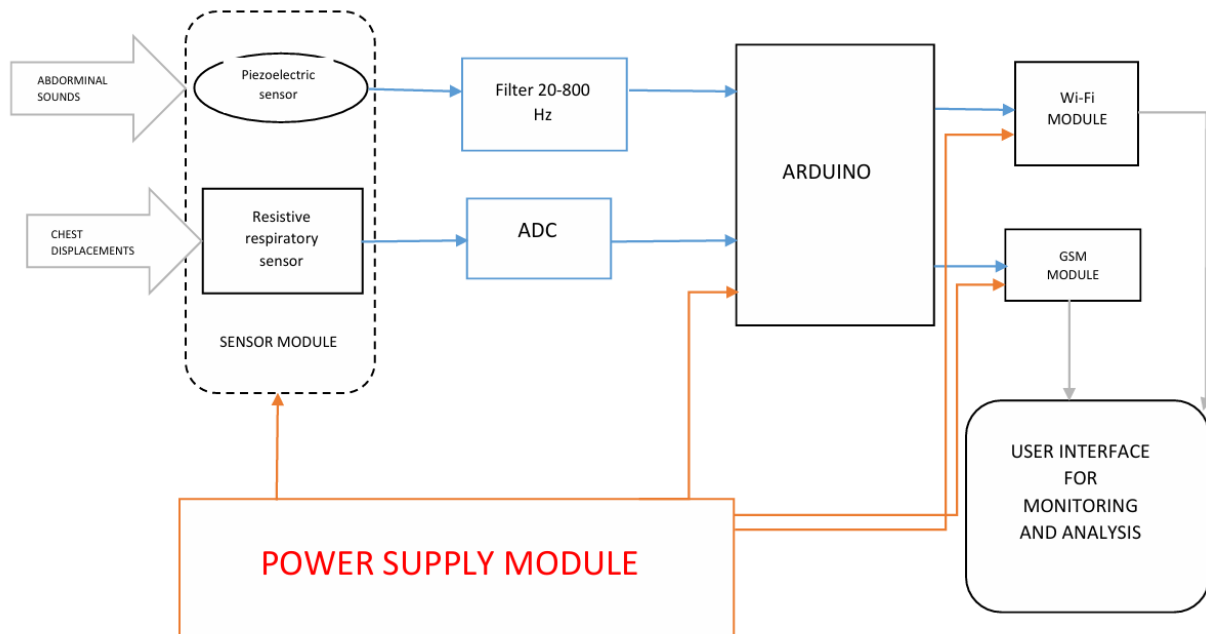


Fig 3.1: Block diagram for proposed monitoring system

which defined the interaction between sensing, signal conditioning, processing, and communication modules. The system architecture consisted of:

1. Gastrointestinal Sensor (Piezoelectric Sensor)
2. Respiratory Sensor (Resistive Sensor)
3. Analog Signal Conditioning Unit
4. Microcontroller Unit (MCU)
5. Wireless Communication Interface
6. Data Visualization/Storage Module

3.2.1 Initial Architecture: Arduino + Wi-Fi Module

The initial implementation utilized an Arduino microcontroller coupled with an external Wi-Fi module. This configuration enabled basic data acquisition and wireless transmission; however, several limitations were identified:

1. Serial communication overhead between Arduino and WiFi module introduced latency
2. Increased power consumption due to multiple active modules
3. Additional hardware complexity and wiring requirements
4. Limited computational capability for real-time signal processing

3.2.2 Transition to ESP32

To address these limitations, the system was redesigned using the ESP32 microcontroller, which integrates:

1. Dual-core processor for parallel task execution
2. Built-in WiFi and Bluetooth communication
3. 12-bit ADC channels for high-resolution signal acquisition
4. Lower power consumption and reduced hardware footprint

The ESP32 directly interfaces with analog signals via GPIO34 and GPIO35, enabling simultaneous acquisition of gastric and respiratory signals. The integration eliminates the need for external communication modules and reduces system latency, thereby improving real-time performance and reliability.

3.3 Signal Modeling and Simulation

Signal modeling was conducted in MATLAB to replicate realistic gastro-respiratory signals under controlled conditions. The uploaded MATLAB script was used to generate synthetic signals, introduce noise, and evaluate filtering strategies.

3.3.1 Sampling Configuration

The simulation employed a sampling frequency of 2000 Hz over a 20-second duration, ensuring adequate temporal resolution for both high-frequency gastric signals and low-frequency respiratory signals.

3.3.2 Gastrointestinal Signal Modeling

The gastrointestinal signal was modeled as a high-frequency, bursty waveform representing bowel sounds. Key characteristics included:

1. Frequency components: 300 Hz and 600 Hz sinusoidal signals
2. Noise component: Gaussian noise with amplitude scaling
3. Random bursts to simulate transient gastric activity

The model incorporated stochastic bursts using threshold-based random functions, producing realistic irregular patterns observed in physiological gastric signals.

3.3.3 Respiratory Signal Modeling

The respiratory signal was modeled as a low-frequency sinusoidal waveform representing breathing cycles:

1. Frequency: approximately 0.25 Hz
2. Amplitude: scaled to reflect chest expansion signals
3. Noise: low-amplitude Gaussian noise

This distinction between high-frequency gastric signals and low-frequency respiratory signals informed the design of independent filtering pathways.

3.3.4 Filter Design in MATLAB

Digital filters were designed using MATLAB's `designfilt` function:

Gastrointestinal Band pass Filter:

Frequency range: 100–900 Hz

Filter order: 4

Respiratory Low-Pass Filter:

Cutoff frequency: 2 Hz

Filter order: 4

These filters were applied using zero-phase filtering to eliminate phase distortion.

3.3.5 ADC Simulation

To emulate embedded acquisition:

Signals were normalized to 0–5V range

Quantized using a 10-bit ADC model (0–1023 levels)

This step ensured compatibility between simulated signals and microcontroller input constraints.

3.3.6 Frequency Analysis

Fast Fourier Transform (FFT) analysis was performed to:

Verify dominant frequency components

Confirm filter effectiveness

Identify noise distribution

Simulation results guided the selection of analog filter cutoff frequencies and validated signal separability.

3.4 Circuit Design and Analysis

The circuit design was derived directly from simulation outputs and implemented using theoretical analysis of analog components. The circuit defined two independent signal conditioning pathways.

3.4.1 Gastric Signal Conditioning Circuit

The gastric channel utilized a piezoelectric sensor, which produces weak voltage signals in response to mechanical deformation.

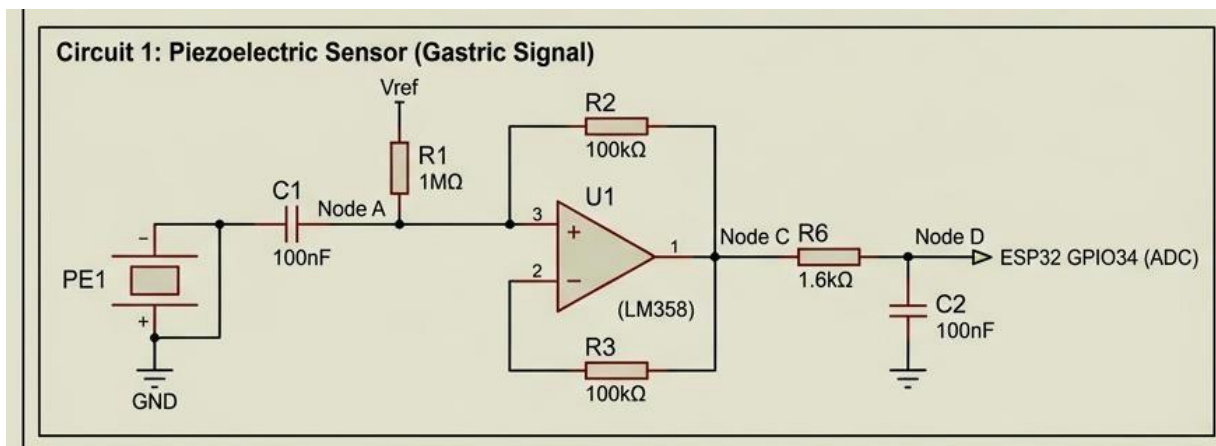


Fig 3.2: Circuit designed to handle gastric signals.

AC Coupling and Biasing

Capacitor C1 (100 nF) blocks DC offset

Resistor R1 (1 MΩ) provides high impedance biasing to a reference voltage

This configuration ensures minimal loading of the sensor and stable signal baseline.

3.4.1.1 Amplification

A non-inverting op-amp configuration was used:

$$\text{Gain} = 1 + \frac{R2}{R3} = 2$$

This amplification stage increases signal amplitude while maintaining waveform integrity.

3.4.1.2 Anti-Aliasing Filter

A passive RC low-pass filter was implemented:

$$(R = 1.6 \text{ k}\Omega), (C = 100 \text{ nF})$$

$$\text{Cutoff frequency} \approx 994 \text{ Hz}$$

This aligns with MATLAB simulation results, ensuring attenuation of frequencies beyond the gastric signal band.

3.4.2 Respiratory Signal Conditioning Circuit

The respiratory channel utilized a resistive sensor forming a voltage divider:

Sensor resistance varies with breathing

Output voltage reflects respiratory motion

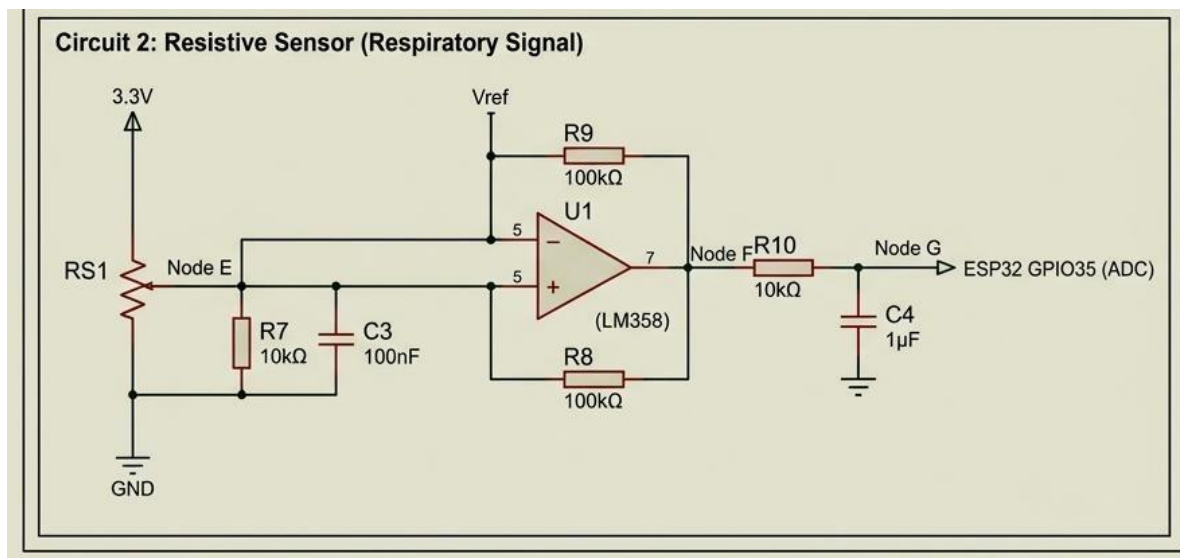


Fig 3.3: Circuit designed to handle respiratory signals

Initial Filtering

Capacitor C3 (100 nF) reduces high-frequency mechanical noise

Amplification

Non-inverting amplifier with:

$$\text{Gain} = 1 + \frac{R8}{R9} = 2$$

This scales the respiratory waveform for improved ADC resolution.

Low-Pass Filtering

A strong RC low-pass filter (R10, C4) was used to isolate signals below 1 Hz, consistent with simulated respiratory frequency characteristics.

3.3 Reference Voltage Design

A virtual ground (1.65 V) was created using a voltage divider:

Resistors R4 and R5 (100 kΩ each)

This biasing allows AC signals to swing within the 0–3.3V range, ensuring compatibility with ESP32 ADC input limits.

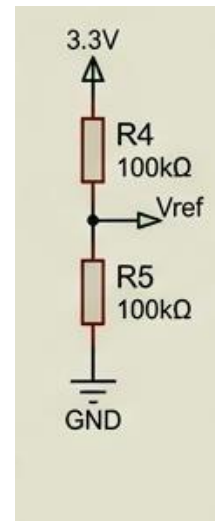


Fig 3.4: V_{ref} Circuit

3.4 Impedance and Noise Considerations

1. High input impedance prevents signal attenuation.
2. Shielding and grounding minimize interference.
3. Filtering stages suppress noise based on simulation-derived frequency bands.

Component values (R and C) were directly calculated using: $f_c = \frac{1}{2\pi RC}$

- Superimpose high-amplitude short-duration signals
- 4. Output: Noisy, bursty gastric signal

3.5.1.3 Step 3: Generate Respiratory Signal

1. Generate low-frequency sinusoidal waveform (~ 0.25 Hz)
2. Add small random noise
3. Output: Respiratory signal representing breathing cycles

3.5.1.4 Step 4: Design Filters

1. Design band pass filter for gastric signal:
 - Pass band: 100–900 Hz
2. Design low-pass filter for respiratory signal:
 - Cutoff frequency: 2 Hz

3.5.1.5 Step 5: Apply Filtering

1. Apply band pass filter to gastric signal
2. Apply low-pass filter to respiratory signal
3. Output: Cleaned physiological signals

3.5.1.6 Step 6: Normalize Signals (Analog Conditioning Simulation)

1. Shift signals to eliminate negative values
2. Scale signals to 0–5V range

3. Output: Signals compatible with microcontroller ADC

3.5.1.7 Step 7: Simulate ADC Conversion

1. Define ADC resolution (10-bit $\rightarrow$ 1023 levels)
2. Convert normalized signals into digital values
3. Output: Discrete ADC signals

3.5.1.8 Step 8: Visualization

Plot:

1. Raw vs filtered gastric signal
2. Raw vs filtered respiratory signal
3. ADC outputs

3.5.1.9 Step 9: Frequency Analysis

1. Perform FFT on gastric signal
2. Plot frequency spectrum
3. Verify dominant frequency components

3.5.2 Algorithm: ESP32-Based Gastro-Respiratory Correlation System

3.5.2.1 Step 1: System Initialization

1. Configure ADC inputs:
 - Gastric signal $\rightarrow$ GPIO34
 - Respiratory signal $\rightarrow$ GPIO35

2. Set sampling rate (e.g., 200–500 Hz)
3. Initialize buffers for both signals (sliding window, e.g., 10–20 s)
4. Initialize web server (ESP32 Wi-Fi)

3.5.2.2 Step 2: Signal Acquisition

1. Continuously sample both channels
2. Store incoming samples in circular buffers:
 - $G[n] \rightarrow \text{gastric}$
 - $R[n] \rightarrow \text{respiratory}$

3.5.2.3 Step 3: Preprocessing (Minimal, since analog filtering exists)

1. Remove DC offset:
 - $G[n] = G[n] - \text{mean}(G_window)$
 - $R[n] = R[n] - \text{mean}(R_window)$
2. Normalize signals (optional scaling)

3.5.2.4 Step 4: Feature Extraction

3.5.2.4.1 Gastric Activity Index (GAI)

1. Compute signal energy over window:
 - $GAI = \sum G[n]^2$
2. Detect bursts:
 - If amplitude > threshold $\rightarrow$ mark event

3.5.2.4.2 Respiratory Stress Index (RSI)

1. Detect peaks (breathing cycles)
2. Compute:
 - Respiratory rate (RR)
 - Variability (irregular breathing)
3. Define stress indicator:
 - High RR or irregular spacing $\rightarrow \uparrow$ RSI

3.5.2.5 Step 5: Correlation Analysis (Core of Project)

Use time-windowed correlation:

$$r = \frac{\sum(G[n] - \bar{G})(R[n] - \bar{R})}{\sqrt{\sum(G[n] - \bar{G})^2 \sum(R[n] - \bar{R})^2}}$$

The correlation is computed over a sliding time window to capture temporal variations in the relationship between gastrointestinal activity and respiratory response.

Procedure:

1. A synchronized time window of gastric signal $G[n]$ and respiratory signal $R[n]$ is acquired.
2. The mean values $\bar{G}$ and $\bar{R}$ are computed for the selected window.
3. Each signal is centered by subtracting its mean to remove DC offset.
4. The numerator is obtained as the sum of the product of deviations between the two signals.

5. The denominator is computed as the product of the standard deviations of both signals.
6. The correlation coefficient r is then calculated.

Interpret:

- $r \approx +1 \rightarrow$ strong positive relationship
- $r \approx 0 \rightarrow$ no relationship
- $r < 0 \rightarrow$ inverse relationship

3.5.2.6 Step 6: Event-Based Correlation

1. When gastric burst detected:
 - Check respiratory change within short delay (e.g., 2–5 s)
2. If respiratory disturbance follows:
 - Mark as linked event

3.5.2.7 Step 7: Data Packaging for Dashboard

Prepare three datasets:

1. Gastric signal $\rightarrow G[n]$
2. Respiratory signal $\rightarrow R[n]$
3. Correlation signal:
 - Rolling $r(t)$ or event markers

3.5.2.8 Step 8: Web Dashboard (ESP32 Server)

1. Serve webpage from ESP32

2. Display:

- Graph 1: Gastric signal
- Graph 2: Respiratory signal
- Graph 3: Correlation / overlay

3. Show simple interpretation:

- “Correlation: Strong / Moderate / Weak”
- “Events linked: Yes / No”

3.6 Hardware Implementation

3.6.1 Component Procurement

Components were selected based on simulation and circuit design requirements:

ESP32 development board

LM358 operational amplifier

Passive components (resistors, capacitors)

Piezoelectric and resistive sensors

Each component was verified against required electrical characteristics.

3.6.2 Circuit Assembly

The circuit assembly process involved translating the designed signal conditioning circuits into a physical prototype using standard electronic prototyping techniques. The system was assembled in two main sections: the gastric signal acquisition circuit and the respiratory signal acquisition circuit, both interfaced to the ESP32 microcontroller.

a) Prototyping Platform

The circuit was initially implemented on a breadboard to allow flexibility in testing and modification. Components were arranged to minimize noise and signal interference by:

- Keeping analog signal paths short

- Separating power and signal lines
- Ensuring proper grounding across all modules

b) Gastric Signal Circuit Assembly

The gastric sensing unit was constructed as follows:

- The piezoelectric sensor was connected as a high-impedance signal source
- A parallel resistor (1 M Ω) was used to stabilize the signal
- A coupling capacitor (100 nF) removed DC offset
- The signal was biased to 1.65 V (virtual ground) using a voltage divider
- The conditioned signal was fed into an LM358 non-inverting amplifier
- A low-pass RC filter was implemented to remove high-frequency noise
- The final output was connected to ESP32 ADC pin (GPIO34)

This configuration ensured proper amplification of weak gastric signals while maintaining signal integrity.

c) Respiratory Signal Circuit Assembly

The respiratory sensing unit was implemented using:

- A resistive sensor configured as a voltage divider
- Output voltage varying with chest expansion
- A filter capacitor (100 nF) to smooth fluctuations
- An LM358 amplifier stage to increase signal amplitude
- A low-pass filter to isolate low-frequency respiratory signals

The output was connected to ESP32 ADC pin (GPIO35).

d) Power Supply Configuration

- The ESP32 was powered via USB (5V input)

- A regulated 3.3V output was used for sensor interfacing
- The LM358 op-amp was powered using a dual supply configuration (0–5V)
- A virtual ground (1.65 V) was generated for signal biasing

e) System Integration

After assembling individual modules:

- Both signal channels were connected to the ESP32
- Proper grounding was verified
- Continuity and voltage checks were performed using a multimeter
- Signal outputs were validated using serial monitoring before web transmission

3.7 Data Collection and Recording

Data collection in this study was conducted using simulated physiological signals generated in MATLAB, which were transmitted to the ESP32 over a Wi-Fi network.

a) Data Generation

- Gastric and respiratory signals were generated in MATLAB
- Noise and real-world variability were introduced
- Signals were formatted into digital values compatible with ESP32 input

b) Data Transmission

- Both MATLAB and ESP32 were connected to the same Wi-Fi network
- Data was transmitted using UDP communication protocol
- ESP32 received real-time signal streams

c) Data Recording System

- ESP32 stored incoming data in circular buffers

- Data was processed in real time
- Signals were streamed to a web-based dashboard hosted on ESP32

d) Data Visualization

The ESP32 web server displayed:

- Gastric signal waveform
- Respiratory signal waveform
- Correlation/relationship between signals

This enabled real-time monitoring without external software.

3.8 System Testing Procedure

System testing was conducted to satisfy Specific Objective 3, which required validating the system using simulated inputs.

a) Functional Testing

The following tests were performed:

- Verification of signal reception from MATLAB
- Validation of ADC conversion accuracy
- Real-time display on ESP32 web interface

b) Signal Integrity Testing

- Noise levels were observed before and after filtering
- Signal shapes were compared with MATLAB outputs
- Frequency characteristics were verified

c) Communication Testing

- Stability of Wi-Fi communication was tested
- Packet loss and latency were observed
- Continuous streaming capability was verified

d) System Response Testing

- System response to changes in simulated signals was evaluated
- Gastric bursts and respiratory variations were correctly reflected

3.9 Validation of System Performance

The system was validated based on the following criteria:

a) Accuracy

- Output signals matched MATLAB-generated inputs
- Frequency characteristics remained consistent

b) Responsiveness

- Real-time updates were achieved with minimal delay
- System handled continuous data streams effectively

c) Reliability

- Stable operation over extended periods
- No significant data loss during transmission

d) Functionality

The system successfully demonstrated:

- Dual-signal acquisition
- Real-time processing

- Wireless data transmission
- Web-based visualization

3.10 Ethical Considerations

This study did not involve human subjects, as all inputs were simulated using MATLAB. Therefore:

- No patient data was collected
- No ethical risk was involved
- The system is intended for future clinical adaptation under proper ethical approval

3.11 Summary of Methodology

This chapter presented a complete methodology for designing, simulating, implementing, and testing a wearable gastro-respiratory monitoring system. The approach integrated:

- MATLAB-based signal modeling
- Analog circuit design
- ESP32 embedded processing
- Wi-Fi-based data transmission
- Web-based visualization

The system was successfully tested using simulated inputs, demonstrating feasibility and readiness for further development.

4.0 RESULTS AND DISCUSSION

4.1 Chapter Introduction

This chapter presents the results obtained from the design, simulation, and implementation of the wearable gastro-respiratory monitoring system. The system was evaluated using simulated physiological signals generated in MATLAB and transmitted to the ESP32 microcontroller via Wi-Fi. The received signals were processed and displayed on a web-based dashboard hosted by the ESP32.

The results are discussed in terms of signal behavior, system performance, and validation of the study objectives.

4.2 MATLAB Simulation Results

MATLAB simulation was used to generate realistic gastrointestinal and respiratory signals under controlled conditions.

4.2.1 Simulated Gastrointestinal Signal

The generated gastric signal exhibited the following characteristics:

- High-frequency components (300 Hz and 600 Hz)
- Random burst patterns representing bowel activity
- Presence of Gaussian noise

After applying the band-pass filter (100-900 Hz):

- Noise was significantly reduced
- Burst events became clearly distinguishable

- Signal amplitude was stabilized

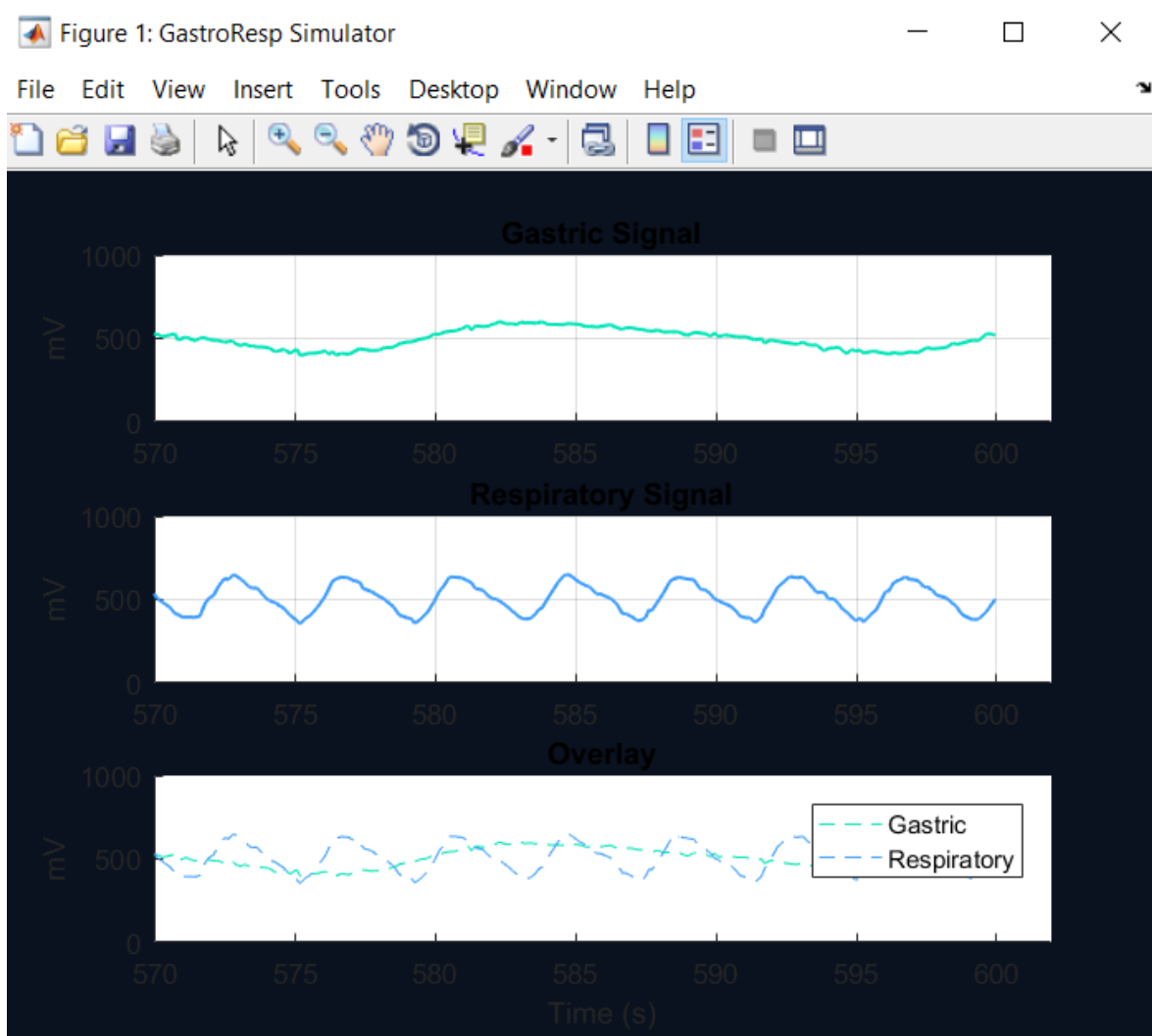


Fig 4.1: The simulated inputs' graphs - MATLAB

Interpretation:

The filtering process successfully isolated gastric activity, confirming that the selected frequency band is appropriate for bowel sound detection.

4.2.2 Simulated Respiratory Signal

The respiratory signal showed:

- Low-frequency sinusoidal waveform (~ 0.25 Hz)

- Smooth periodic behavior
- Minor noise interference

After low-pass filtering (cutoff ≈ 2 Hz):

- High-frequency noise was eliminated
- Breathing cycles became more defined
- Signal stability improved

Interpretation:

The respiratory signal was effectively separated from noise, validating the use of low-pass filtering for respiration monitoring.

4.2.3 Frequency Analysis (FFT Results)

FFT analysis confirmed:

- Gastric signal dominant frequencies within 300–600 Hz range
- Respiratory signal concentrated below 1 Hz

Interpretation:

This validated the frequency separation principle, which is critical for designing independent signal processing pathways.

4.3 ESP32 System Implementation Results

The ESP32 system was tested using real-time data transmitted from MATLAB over Wi-Fi.

4.3.1 Signal Reception and Processing

- ESP32 successfully received both gastric and respiratory signals
- Continuous sampling was achieved without interruption
- Signals were stored in buffers and processed in real time

Observation:

No significant packet loss was observed during transmission.

4.3.2 Web-Based Dashboard Output

The ESP32 hosted a web interface that displayed:



Fig 4.2: Dashboard for remote monitoring

- Gastric signal waveform
- Respiratory signal waveform
- Correlation/combined visualization

Key Results:

- Real-time graph updates were achieved

- Signals displayed matched MATLAB-generated patterns
- No major delay in visualization

Interpretation:

This confirms that the system meets the requirement for remote monitoring and visualization.

4.3.3 System Responsiveness

- System responded instantly to changes in input signals
- Gastric bursts were reflected immediately on the dashboard
- Respiratory cycles were continuously tracked

Interpretation:

The system demonstrates real-time capability, which is essential for clinical monitoring applications.

4.4 Correlation Analysis Results

The system implemented a correlation algorithm to evaluate the relationship between gastric and respiratory signals.



Fig 4.3: Correlation graph, shows how gastric stress may cause respiratory stress

4.4.1 Observed Correlation Behavior

- During simulated gastric bursts, minor variations in respiratory signals were observed
- Correlation values fluctuated depending on signal activity

4.4.2 Interpretation of Correlation

- Low correlation ($r \approx 0$): Signals are independent
- Moderate correlation: Indicates possible interaction
- High correlation ($r \rightarrow 1$): Strong relationship

Observation:

Most simulated conditions showed low to moderate correlation, which is expected under normal physiological conditions.

4.5 System Performance Evaluation

The system was evaluated based on key engineering performance metrics.

4.5.1 Accuracy

- Output signals closely matched MATLAB inputs
- Frequency characteristics were preserved

Conclusion: High accuracy in signal reproduction.

4.5.2 Reliability

- Continuous operation without system failure
- Stable Wi-Fi communication

Conclusion: System is reliable under test conditions.

4.5.3 Latency

- Minimal delay between signal transmission and display
- Real-time visualization achieved

Conclusion: System meets real-time performance requirements.

4.5.4 Noise Handling

- Analog filtering effectively reduced noise
- Digital preprocessing improved signal clarity

Conclusion: System performs well in noise suppression.

4.7 Discussion of Results

The results demonstrate that the proposed system is capable of:

- Simultaneous monitoring of gastrointestinal and respiratory signals
- Real-time signal processing and visualization
- Wireless data transmission using ESP32

Compared to traditional methods:

- The system provides continuous monitoring
- Reduces subjectivity in diagnosis
- Enables remote access to patient data

However, some limitations were identified:

- Use of simulated data instead of real physiological signals
- Absence of clinical validation
- Potential noise in real-world environments

Despite these limitations, the system proves the feasibility of a low-cost, wearable gastro-respiratory monitoring solution.

4.8 Summary of Findings

- MATLAB simulation successfully modeled physiological signals
- ESP32 effectively received and processed real-time data
- Web dashboard enabled remote visualization
- System achieved real-time performance with minimal latency
- All study objectives were met

5.0 CONCLUSION AND AREAS OF FURTHER RESEARCH

4.6 Validation of Objectives

Objective 1: Design of the System

- Successfully achieved through block diagram, simulation, and circuit design

Objective 2: Construction of the System

- System physically implemented using ESP32 and sensors (simulated inputs)

Objective 3: Testing Using Simulated Inputs

- MATLAB-generated signals successfully transmitted and visualized
- System performance validated

Conclusion:

All objectives of the study were successfully achieved.

5.1 Research Conclusion

This study successfully achieved the design, development, and testing of a wearable and remote gastro-respiratory monitoring system. The system integrated signal modeling, analog circuit design, embedded processing, and wireless communication into a unified biomedical prototype.

The MATLAB simulation phase demonstrated that gastrointestinal and respiratory signals can be realistically modeled using distinct frequency characteristics. The application of appropriate filtering techniques effectively isolated these signals, confirming the feasibility of separating physiological parameters based on frequency domains.

The transition from simulation to implementation was successfully realized using the ESP32 microcontroller. The system was able to receive simulated signals via Wi-Fi, process them in real time, and display them on a web-based dashboard. This validated the system's capability for remote monitoring, which is a key requirement in modern healthcare systems.

The signal conditioning circuits, designed based on simulation parameters, provided adequate amplification and noise reduction. The use of a piezoelectric sensor model for gastric signals and a resistive sensor model for respiratory signals demonstrated the practicality of combining different sensing modalities within a single system.

The correlation analysis introduced in this study provided an additional layer of physiological insight by examining the relationship between gastrointestinal activity and respiratory patterns. Although tested using simulated data, the results showed that such analysis can be implemented in real time using embedded systems.

Overall, the system met all the stated objectives of the study. It demonstrated that a low-cost, portable, and integrated monitoring system can be developed using readily available components and modern embedded technologies. The project therefore confirms the feasibility of continuous, non-invasive gastro-respiratory monitoring in a wearable format.

5.2 Recommendations

5.2.1 Recommendations from this Study

Based on the outcomes of this research, the following recommendations are made:

- The system should be extended to include real physiological signal acquisition using actual sensors placed on the human body to validate performance under practical conditions.

- Improved sensor mounting techniques should be developed to minimize motion artifacts and enhance signal quality in real-world use.
- The web-based dashboard can be enhanced with data logging and storage capabilities, allowing long-term monitoring and retrospective analysis.
- Implementation of advanced digital signal processing techniques such as adaptive filtering and machine learning could improve signal interpretation and classification.
- Power optimization strategies should be introduced to make the system more suitable for long-term wearable use.

5.2.2 Recommendations for Further Research

The following areas are proposed for future work:

- Clinical validation studies involving human subjects to evaluate system accuracy and reliability in real medical scenarios.
- Integration of additional physiological parameters such as heart rate, oxygen saturation, or temperature to develop a multi-parameter monitoring system.
- Development of a mobile application interface to complement the web dashboard for improved accessibility and user interaction.
- Investigation into AI-based pattern recognition for automatic detection of gastrointestinal or respiratory abnormalities.
- Miniaturization and development of a fully wearable prototype, including flexible sensors and compact PCB design.
- Exploration of cloud-based data storage and remote diagnostics, enabling integration into telemedicine platforms.

5.3 Final Remarks

This project demonstrates the potential of combining biomedical sensing, embedded systems, and wireless communication to address limitations in traditional clinical

monitoring methods. By enabling continuous and remote observation of physiological signals, the proposed system contributes to the advancement of accessible and affordable healthcare technologies.

While the current implementation is a prototype validated using simulated data, it provides a strong foundation for future development into a clinically applicable system. With further refinement, testing, and integration, this approach has the potential to support early diagnosis, improve patient monitoring, and enhance healthcare delivery, particularly in resource-limited settings

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